

P0591r1 | Utility functions to implement uses-allocator construction

Pablo Halpern phalpern@halpernwrightsoftware.com

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1 Abstract

The phrase “*Uses-allocator construction* with allocator `Alloc`” is defined in section `[allocator.uses.construction]` of the standard (20.7.7.2 of the 2014 standard or 20.10.7.2 of the 2016 CD). Although the definition is reasonably concise, it fails to handle the case of constructing a `std::pair` where one or both members can use `Alloc`. This omission manifests in significant text describing the `construct` members of `polymorphic_allocator` `[memory.polymorphic_allocator.class]` and `scoped_allocator_adaptor` `[allocator.adaptor]`. Additionally, a `vector<pair<T,U>, A>` fails to pass the allocator to the pair elements if `A` is a scoped or polymorphic allocator.

Though we could add the `pair` special case to the definition of *Uses-allocator construction*, the definition would no longer be concise. Moreover, any library implementing features that rely on *Uses-allocator construction* would necessarily centralize the logic into a function template. This paper, therefore, proposes a set of templates that do exactly this centralization, in the standard. The current uses of *Uses-allocator construction* could then simply defer to these templates, making those features simpler to describe and future-proof against other changes.

Because this proposal modifies wording in the standard, it is targeted at C++20 (aka, C++Next) rather than a technical specification.

2 Changes from R0

- Fixed function template prototypes, which incorrectly depended on partial specialization of functions.

3 Choosing a direction

Originally, I considered proposing a pair of function templates, `make_using_allocator<T>(allocator, args...)` and `uninitialized_construct_using_allocator(ptrToT, allocator, args...)`. However, implementation experience with the feature being proposed showed that, given a type `T`, an allocator `A`, and an argument list `Args...`, it was convenient to generate a `tuple` of the final argument list for `T`’s constructor, then use `make_from_tuple` or `apply` to implement the above function templates. It occurred to me that exposing this `tuple`-building function may be desirable, as it opens the door to an entire category of functions that use `tuples` to manipulate argument lists in a composable fashion.

If the basics of this proposal are accepted by LEWG, there would need to be a discussion of exactly what should be standardized. The options are:

1. Standardize the function template that generates a `tuple` of arguments.

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2. Standardize the function templates that actually construct a T from an allocator and list of arguments.
 3. Both.

This proposal chooses option 3, but I am open to the other options.

4 Proposed wording

The following wording is still rough. More detailed wording to come after LEWG review and revision. Wording is relative to the November 2016 Committee Draft, N5131.

Guidance needed: The wording uses `forward_as_tuple`, which prevents copies, and doesn't require copy- or move-constructibility, but can result in dangling references if the resulting `tuple` outlives the full expression in which it was created. Is this OK? If so, should I repeat the cautionary words already found in the description of `forward_as_tuple`?

Add the following new function templates to `<memory>`:

```
template <class T, class Alloc, class... Args>
    auto uses_allocator_construction_args(const Alloc& a, Args&&... args) -> see below;
```

Remark: T is not deduced and must therefore be specified explicitly by the caller. This template does not participate in overload resolution if T is a specialization of `std::pair`.

Returns: A `tuple` value determined as follows:

- if `uses_allocator_v<T, Alloc>` is false and `is_constructible_v<T, Args...>` is true, return `forward_as_tuple(std::forward<Args>(args)...)...`.
- otherwise, if `uses_allocator_v<T, Alloc>` is true and `is_constructible_v<T, allocator_arg_t, Alloc, Args...>` is true, return `forward_as_tuple(allocator_arg, alloc, std::forward<Args>(args)...)...`.
- otherwise, if `uses_allocator_v<T, Alloc>` is true and `is_constructible_v<T, Args..., Alloc>` is true, return `forward_as_tuple(std::forward<Args>(args)..., alloc)`.
- otherwise, the program is ill-formed. [*Note:* An error will result if `uses_allocator_v<T, Alloc>` is true but the specific constructor does not take an allocator. This definition prevents a silent failure to pass the allocator to a constructor. — *end note*]

```
template <class T, class... Args1, class... Args2>
    auto uses_allocator_construction_args(const Alloc& a,
                                           piecewise_construct_t,
                                           tuple<Args1...> x,
                                           tuple<Args2...> y) -> see below;
```

Remark: T is not deduced and must therefore be specified explicitly by the caller. This template does not participate in overload resolution unless T is a specialization of `std::pair`.

Returns: For T specified as `pair<T1, T2>`, equivalent to

```

make_tuple(piecewise_construct,
    apply(x, [](Args1... args1) -> auto {
        return uses_allocator_construction_args<T1>(a,
            std::forward<Args1>(args1)...);
    }),
    apply(y, [](Args2... args2) -> auto {
        return uses_allocator_construction_args<T2>(a,
            std::forward<Args2>(args2)...);
    }));

```

```

template <class T>
auto uses_allocator_construction_args(const Alloc& a) -> see below;

```

Remark: T is not deduced and must therefore be specified explicitly by the caller. This template does not participate in overload resolution unless T is a specialization of `std::pair`.

Returns: For T specified as `pair<T1, T2>`, equivalent to `uses_allocator_construction_args<pair<T1,T2>>(a, piecewise_construct, tuple<>{}, tuple<>{})`

```

template <class T, class U, class V>
auto uses_allocator_construction_args(const Alloc& a, U&& u, V&& v) -> see below;

```

Remark: T is not deduced and must therefore be specified explicitly by the caller. This template does not participate in overload resolution unless T is a specialization of `std::pair`.

Returns: For T specified as `pair<T1, T2>`, equivalent to `uses_allocator_construction_args<pair<T1,T2>>(a, piecewise_construct, forward_as_tuple(std::forward<U>(u)), forward_as_tuple(std::forward<V>(v)))`.

```

template <class T, class U, class V>
auto uses_allocator_construction_args(const Alloc& a, const pair<U,V>& pr) -> see below;

```

Remark: T is not deduced and must therefore be specified explicitly by the caller. This template does not participate in overload resolution unless T is a specialization of `std::pair`.

Returns: For T specified as `pair<T1, T2>`, equivalent to `uses_allocator_construction_args<pair<T1,T2>>(a, piecewise_construct, forward_as_tuple(pr.first), forward_as_tuple(pr.second))`.

```

template <class T, class U, class V>
auto uses_allocator_construction_args(const Alloc& a, pair<U,V>&& pr) -> see below;

```

Remark: T is not deduced and must therefore be specified explicitly by the caller. This template does not participate in overload resolution unless T is a specialization of `std::pair`.

Returns: For T specified as `pair<T1, T2>`, equivalent to `uses_allocator_construction_args<pair<T1,T2>>(a, piecewise_construct, forward_as_tuple(std::forward<U>(pr.first)), forward_as_tuple(std::forward<V>(pr.second)))`.

```

template <class T, class Alloc, class... Args>
T make_using_allocator(const Alloc& a, Args&&... args);

```

Remark: T is not deduced and must therefore be specified explicitly by the caller. This template does not participate in overload resolution unless T is a specialization of `std::pair`.

Returns: For T specified as `pair<T1, T2>`, equivalent to

```
make_from_tuple<T>(
    uses_allocator_construction_args<T>(a, forward<Args>(args)...));
```

```
template <class T, class Alloc, class... Args>
T* uninitialized_construct_using_allocator(T* p,
                                           const Alloc& a,
                                           Args&&... args);
```

Remark: T is not deduced and must therefore be specified explicitly by the caller. This template does not participate in overload resolution unless T is a specialization of `std::pair`.

Returns: For T specified as `pair<T1, T2>`, and given the exposition-only function template:

```
template <class T, class... A>
uninitialized_construct_from_tuple(T* p, tuple<A...&& t) {
    apply(std::move(t), [](A&&... args) {
        ::new(static_cast<void*>(p)) T(std::forward<A>(args)...);
    });
}
```

equivalent to

```
uninitialized_construct_from_tuple(
    p,
    uses_allocator_construction_args<T>(a, forward<Args>(args)...));
```

Guidance Needed: Should we consider adding `uninitialized_construct_from_tuple` as a separate (non-exposition) function, since it appears to be useful and it's hard to do the same thing without creating a named function?

Additionally, rewrite the `construct` methods of `polymorphic_allocator` and `scoped_allocator_adaptor` in terms of the above.

Consider replacing all uses of *uses allocator construction* with references to these functions and removing *uses allocator construction* from the standard.